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Project #042: Reduction of hip fractures after elective total hip arthroplasty within 90 days

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Abstract

The rate of post operative hip fractures at HF Wyandotte was above the average across the state based on data from the Michigan Arthroplasty Registry Collaborative Quality Initiative (MARCQI) for BCBS.

Hip fractures after surgery cause dissatisfaction for patients due to limited mobility and pain. They also create a financial burden for hospitals.

Our goal was to decrease hip fractures below 1.05% based on the MARCQI goal. Our data was tracked through the MARCQI database.

When we started our rate was 5.10% from 3/1/23-3/1/24.

To decrease our hip fractures our primary hip surgeon started cementing femoral stems in females older than 70 years based on recommended data due to poor bone quality in older females. This started in March 2024.

Current data from 3/1/24-3/1/25 shows our rate at 0.81%.

Plan / Introduction

Problem Description: Hip fractures after elective total hip arthroplasties

Available Knowledge: Hip fractures after total hip arthroplasties are more common in women over 70 years old.

Rationale: Previous studies have shown a decrease in hip fractures among women when cemented stems are used.

Specific Aim: Our aim was to decrease hip fractures below 1.05%.

Do / Methods

Intervention: To decrease our hip fractures our primary hip surgeon started cementing femoral stems in females older than 70 years.

Study of the Intervention: Data was compared prior to the intervention and after the intervention. Data showed a significant decrease in hip fractures after the intervention was implemented.

Measures: Data looked at hip fractures within this time frame including female and male. Data included hip fractures for our primary hip surgeon. Hip fractures included required a change in treatment plan per MARCQI definition (fixation, change in weight bearing, intention to monitor with increased follow up or orders for braces in relation to fracture).

Analysis: Data showed improvement in our hip fractures. Data can vary based on gender and surgeon. Our targeted group was females, while data does show hip fractures in male patients. Our targeted group also included our primary hip surgeon. Other surgeon's perform hip surgeries that may have hip fractures. Data for other surgeons showed one additional fracture. Future data shows one fracture for primary surgeon.

Check / Results

Our primary hip surgeon started cementing femoral stems in females over 70 in March of 2024.

Data from March 1, 2023 to March 1, 2024 showed 5 fractures out of 98 elective total hip arthroplasties. 4 of these were female and one was male. (See Figure 1)

Data from March 1, 2024 to March 1, 2025 showed 1 fracture out of 124 elective total hip arthroplasties. This fracture was in a male patient. (See Figure 2)

Data did not include hip fractures from other surgeons.

Cementing was done outside the age range when surgeon deemed appropriate.

Data

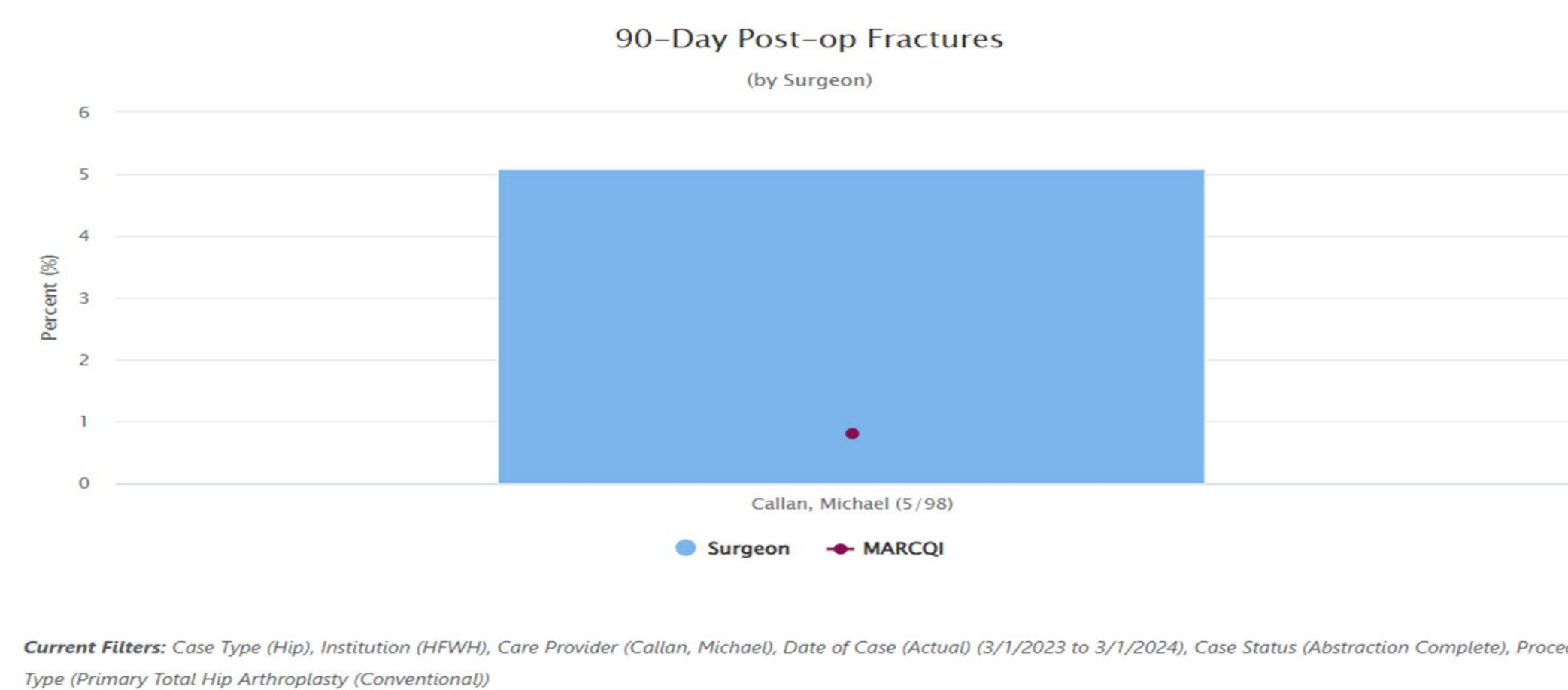


Figure 1

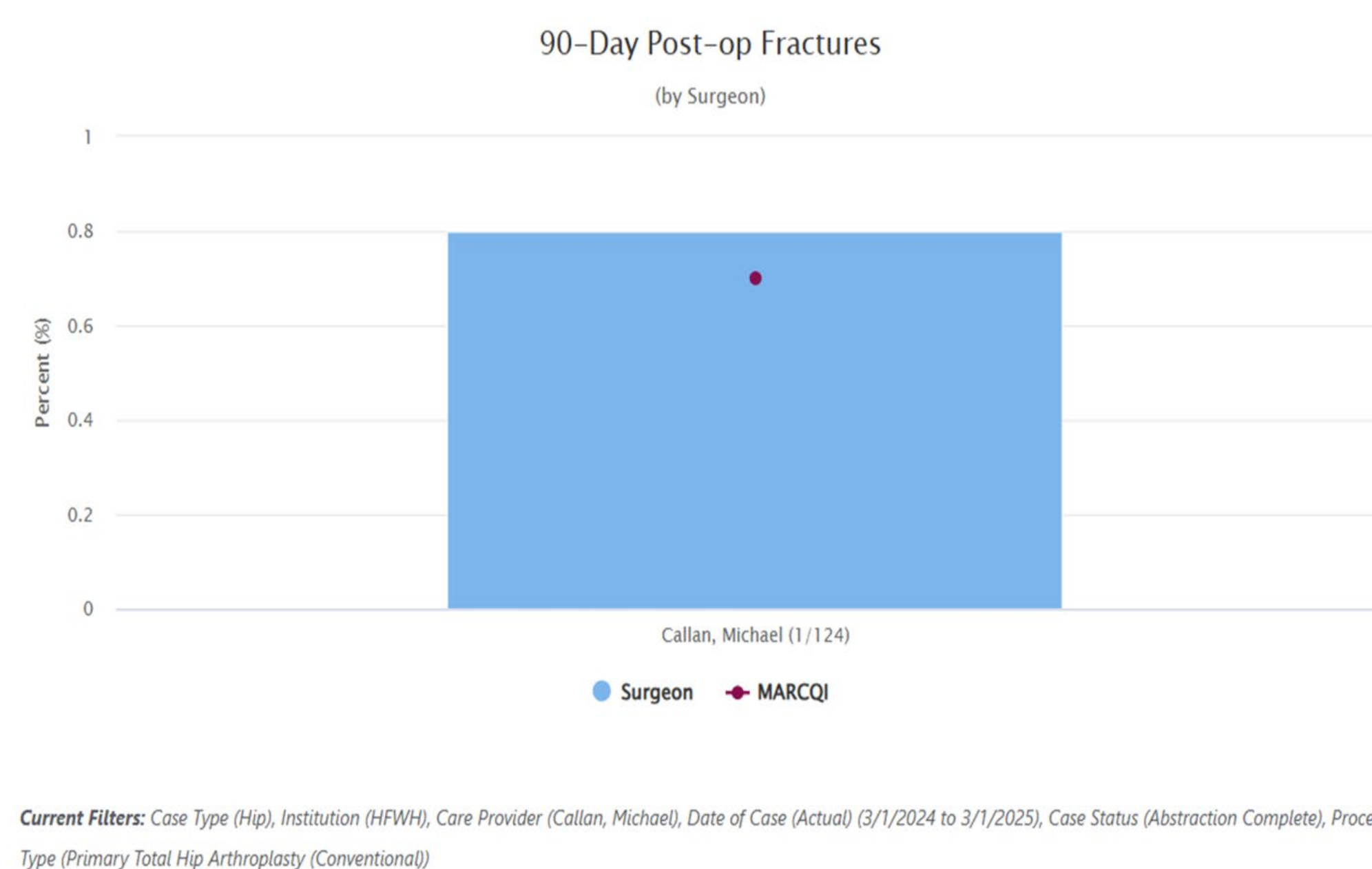


Figure 2

Act / Discussion

Summary: The aim of this was to decrease hip fractures. Key findings indicate that cementing hips in females over 70 years old can decrease the fracture rate.

Interpretation: Cementing stems in total hip arthroplasty is a practice change. Previous studies show an overall benefit in cementing stems in elderly females to prevent fractures. Overall, this can result in increased patient satisfaction and decreased cost to hospitals with less complications.

Limitations: Limits to this were that not all surgeons were included and fractures that did not require treatment were excluded. Data on cementing was included over one year's time. Another factor was the fracture rate in male patients. Finally, the age when patients are cemented can vary.

Conclusions: Overall, this showed cementing stems in female patients over 70 can decrease the fracture rate. Using data from the MARCQI database was a useful way to track data. Fracture rates will be tracked in the future to monitor rates.

Other Information

Acknowledgements: Data was available through the MARCQI database through BCBS.

Funding: Cement was an additional cost provided by the hospital.

Ethical Considerations: There were no ethical considerations related to the data.

References

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